

Lumbini City College

Send UP Exams 2026

Bachelor In Computer Application

Course Title: Microprocessor & Computer Architecture

Code No: CACS 155

Time: 3 hours

Semester: II

Full Marks: 60

Pass Marks: 24

Candidates are required to answer the question in their own words as far as possible.

SET A

Group A

Attempt all questions.

10 × 1 = 10

1. What is a microprocessor?
2. What is an ALU?
3. What is an opcode?
4. What is an addressing mode?
5. What is a micro-operation?
6. What is a microinstruction?
7. What is RISC?
8. What is Booth multiplication?
9. What is DMA?
10. What is pipelining?

Group B

Attempt any FIVE questions.

6 × 5 = 30

11. Explain the functional block diagram of 8085.
12. Explain the registers and flag register of 8085.
13. Explain the different addressing modes of 8085 with examples.
14. Explain the different types of 8085 instructions with examples.
15. Write an 8085 program to add two 8-bit numbers.
16. Write an 8085 program to subtract two 8-bit numbers.

Group C

Attempt any TWO questions.

2 × 10 = 20

17. Explain the architecture of 8085 with a neat block diagram. Explain the functions of its major components.
18. Explain microprogrammed control unit, control memory, address sequencing and microinstruction format.
19. Explain pipelining, arithmetic pipeline, instruction pipeline and pipeline hazards.

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SET B

Group A

Attempt all questions.

10 × 1 = 10

1. List the main components of a microprocessor.
2. What is a system bus?
3. What is a flag register?
4. What is a machine cycle?
5. What is an interrupt?
6. What is a common bus system?
7. What is stack organization?
8. What is 2's complement?
9. What is cache memory?
10. What is a pipeline hazard?

Group B

Attempt any FIVE questions.

6 × 5 = 30

11. Explain the pin configuration of 8085.
12. Explain arithmetic and logical instructions of 8085 with examples.
13. Write an 8085 program to multiply two 8-bit numbers.
14. Write an 8085 program to divide two 8-bit numbers.
15. Explain arithmetic, logic and shift micro-operations with examples.
16. Explain 3-address, 2-address, 1-address and 0-address instructions.

Group C

Attempt any TWO questions.

2 × 10 = 20

17. Explain the 8085 instruction set with suitable examples. Also explain the different types of addressing modes.
18. Explain RISC and CISC architectures. Compare their major characteristics.
19. Explain I/O organization, DMA, memory hierarchy, RAM, ROM and cache memory.